

There are several possible ways to achieve this adjustment and most good market data applications offer a choice in this regard, but it does not make a huge difference for the bigger picture which exact method you choose. My preferred method is to identify the liquid contract based on open interest and to link the contracts together so that the old contract's close matches the new contract's close on the rollover date, keeping any actual gaps on that day and back-adjusting the entire time series all the way from the start of the data. So if you look at the adjusted time series to see the exact price of corn in June of 1985, it will be very different from what the actual price was at the time since there have been countless adjustments done with all the roll-overs that have occurred since. However, the real trends of the price series over time have been properly preserved and the most recent price in the series represents the actual price in the market.

Other methods of adjusting prices can involve using ratios of two contracts, forward adjusting, and different methods of using volume and open interest to find the most liquid issue. These are details which are not too important for the long-term strategy but worth experimenting with if you want to look at the finer details. For a comprehensive look at rollover methods, see Jack Schwager's (1995) book, *Schwager on Futures: Technical Analysis*.

The Change in How Continuations Are Used

The obvious problem with these calculated continuations is that they are not real. They are not tradable instruments, and they do not represent actual, real price moves. They are merely rough estimates of what a long-term holding would have looked like and they are just calculated and highly theoretical time series. If you need long-term analytics, such as calculating a one hundred-day moving average, you absolutely need continuations. Doing this sort of maths on individual futures contracts makes little sense.

The more important question is if and how continuations should be used in back-testing strategies. Traditionally, most people doing such back-testing used these pre-calculated series for the simulated trading. It's a significant simplification of reality but it serves as a good-enough approximation.

The issue with this is not just that the price level will be different, obfuscating the number of contracts required, the margin requirements and such, but also that it takes rolling and roll cost out of the picture.

If you apply trend-following strategies on continuations, it will appear as if you could simply buy and hold oil or gold or corn for years and years. The rolling is an added complication in real life, and, if possible you should try to include this in the backtest logic.